UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 26: Fourier Applications

Reference: Chapter 6
(Section 6.11, 6.24, 6.25)

Important Reminder

- Please remember to complete your online course evaluation on https://accessUH.uh.edu

Faculty/Course Evaluation

- Your candid feedback will help us further improve ECE 3337

Final Exam

Date: May 11th 2 - 4pm, SW 102 (Social Work Building).

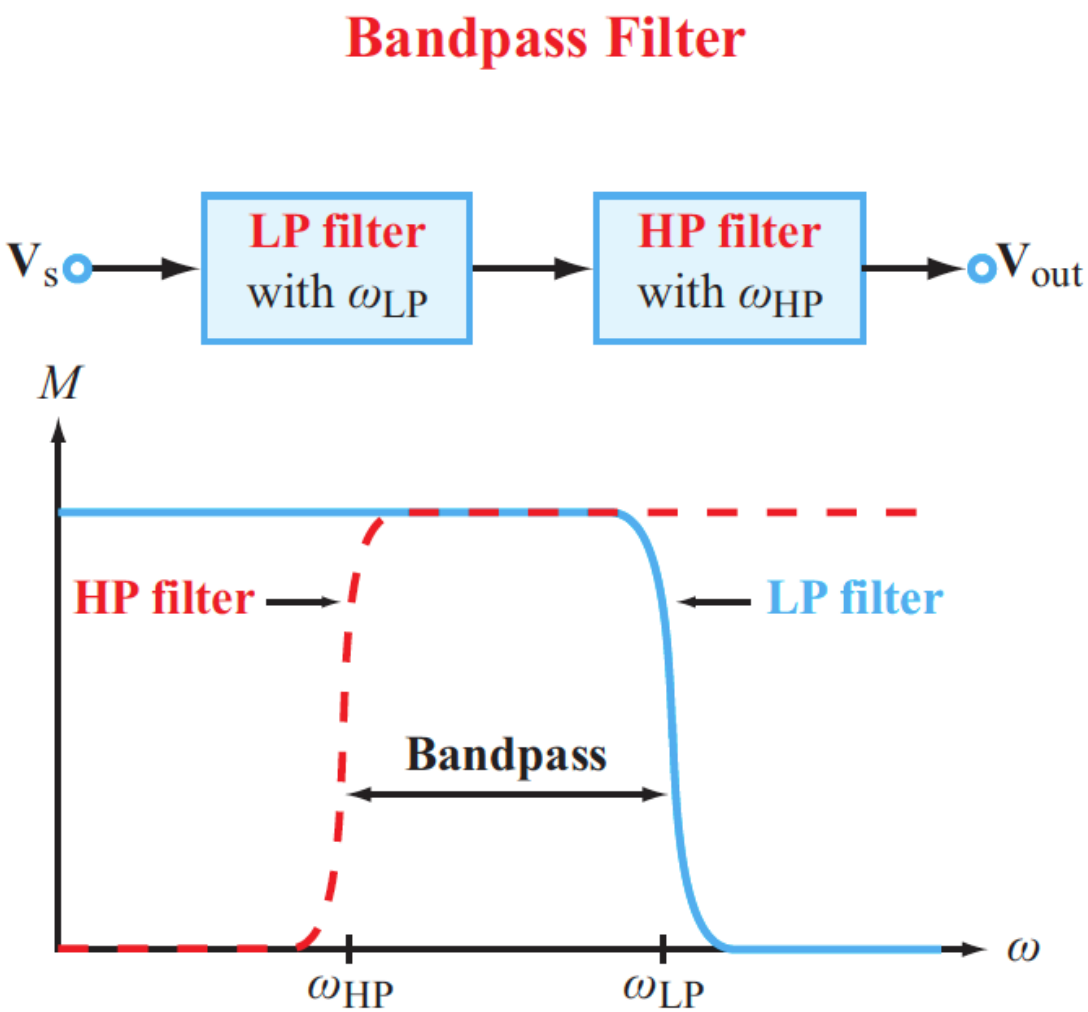
https://www.uh.edu/infotech/services/facilities-equipment/supported-classrooms/sw/102/

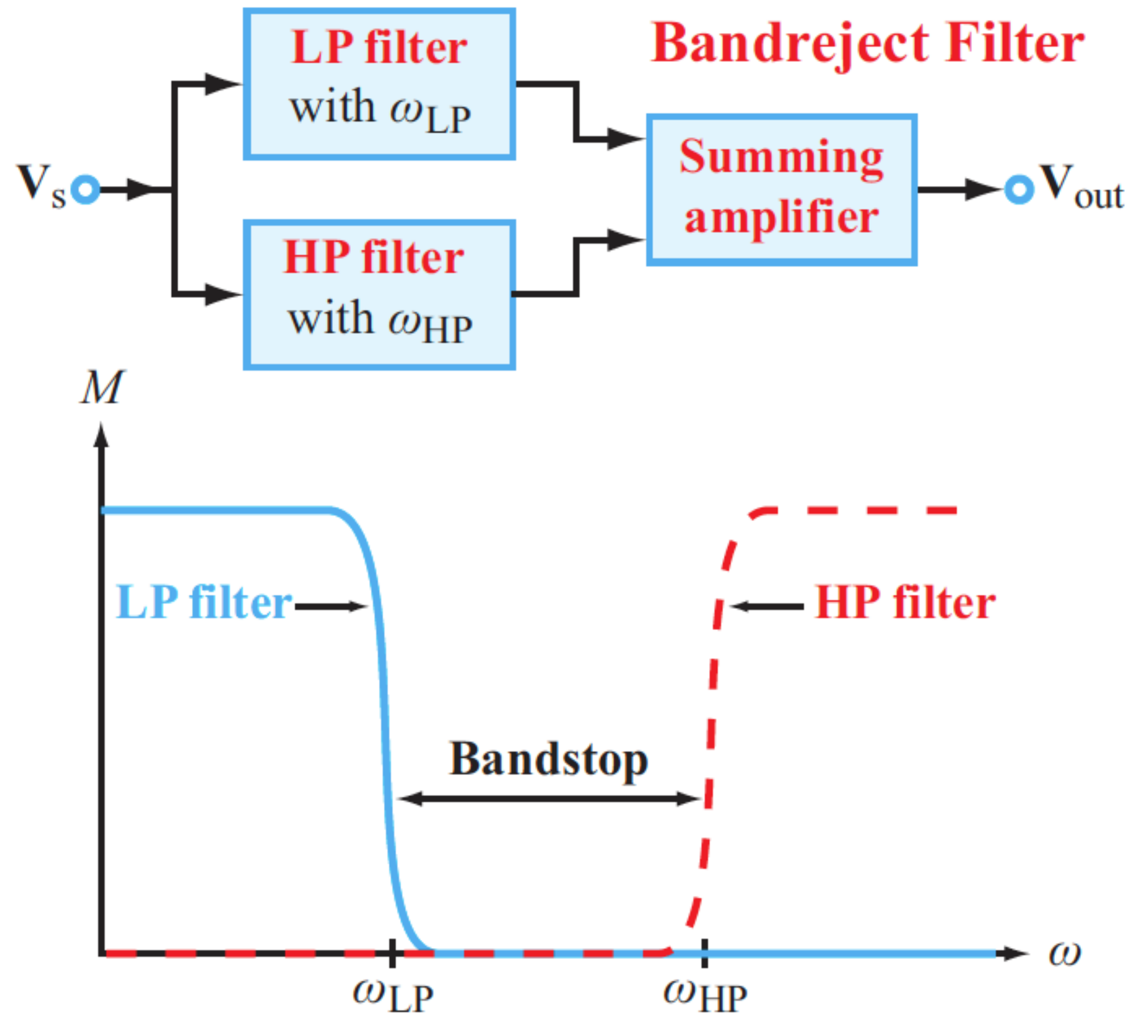
Plenty of old exams are posted on MS Teams for practice

- Practice for accuracy and speed
- Test yourself individually, rather than as a group
- Avoid a false sense of knowing a topic
  - This can easily happen in a group setting
- Suggested study strategy:
  - Do the practice tests individually
  - Identify the challenging problems, and ask your instructors for help
  - Discuss the challenging problems in your study group
  - Follow up with individual work to confirm your understanding

Recap: Combining Filters

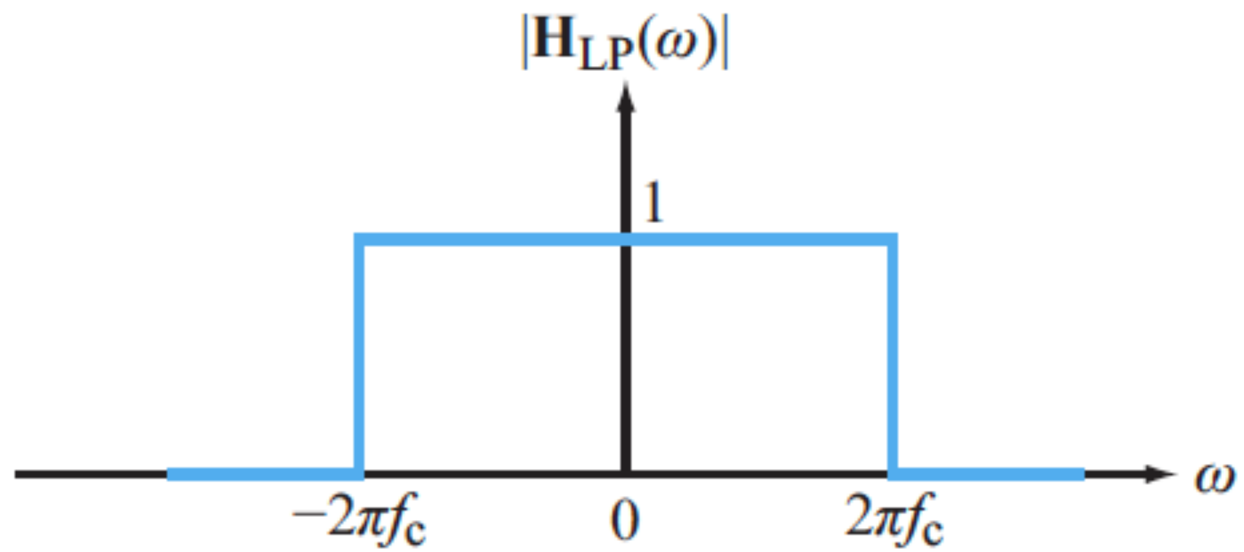
Basic filters can be combined in series and/or parallel to build more complex or customized filters





Ideal Low-pass Filter Revisited

- Until now, we have plotted the frequency response of filters as a function of only positive frequencies
- Even though negative frequencies have no physical meaning, it is mathematically convenient to define ω over $-\infty$ to ∞, just as we do with time t.

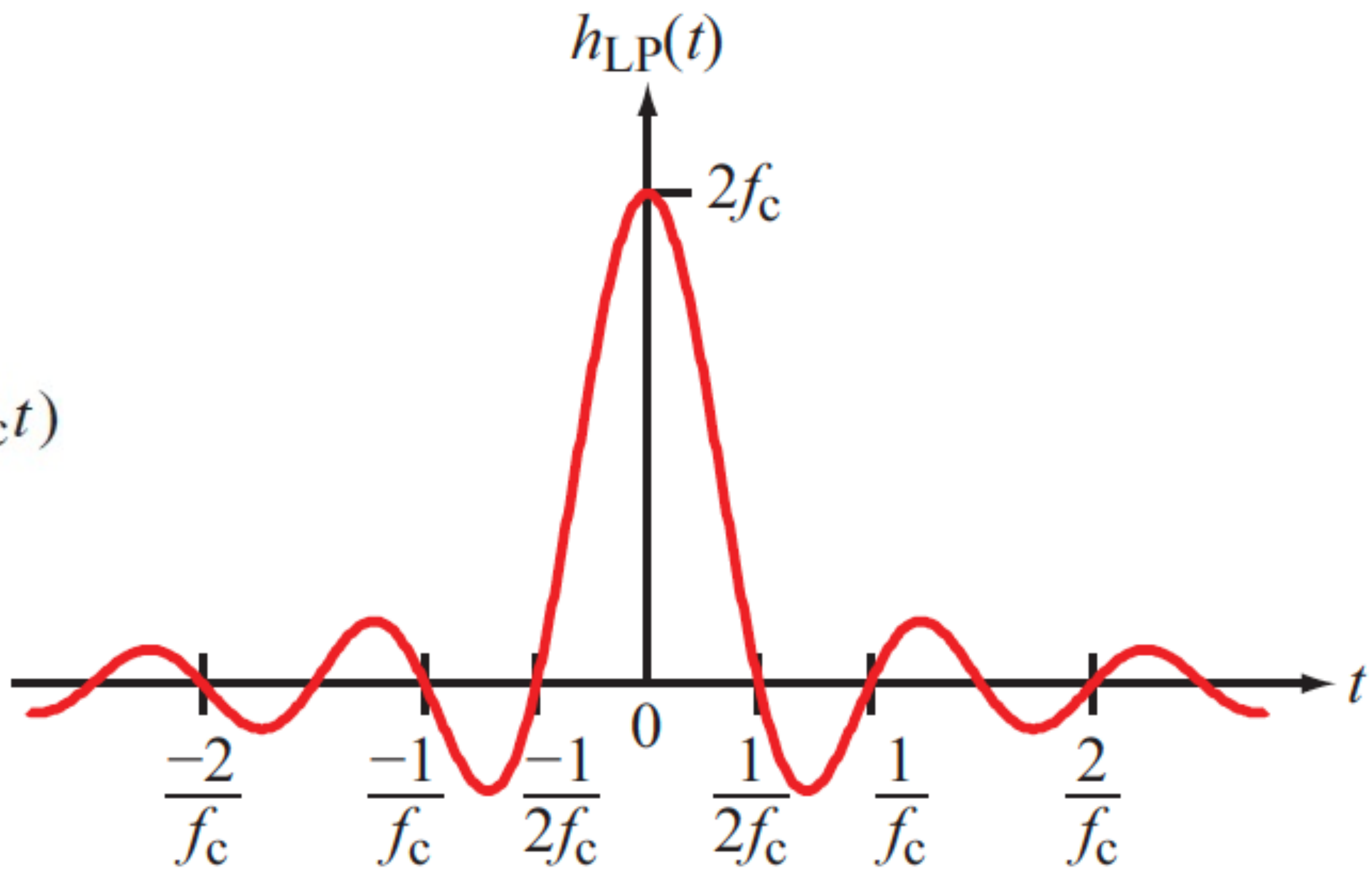


“Brick-wall” Low-pass Filter

Ideal Low-pass Filter in the Time Domain

- **The impulse response of the brick-wall filter**

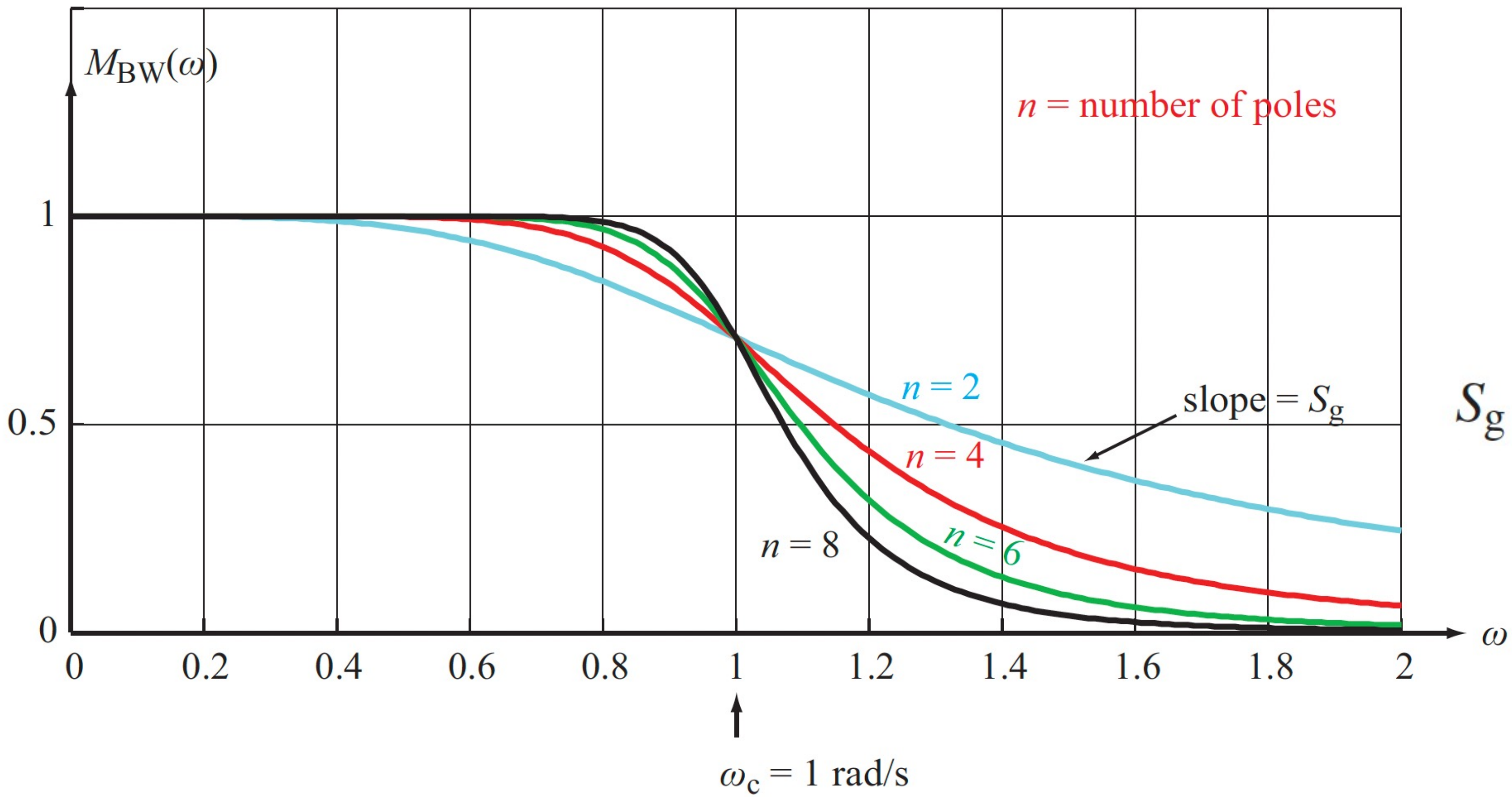
$$h_{\mathrm{LP}}(t) = \mathcal{F}^{-1}\{\mathbf{H}_{\mathrm{LP}}(\omega)\}$$
$$= \frac{\sin(2\pi f_c t)}{\pi t} = 2 f_c \,\mathrm{sinc}(2\pi f_c t)$$



“Brick-wall” Low-pass Filter

- **This is noncausal, so such a low-pass filter does not exist physically**
- **However, we can approximate it well enough to be useful**

Example: Butterworth Filters



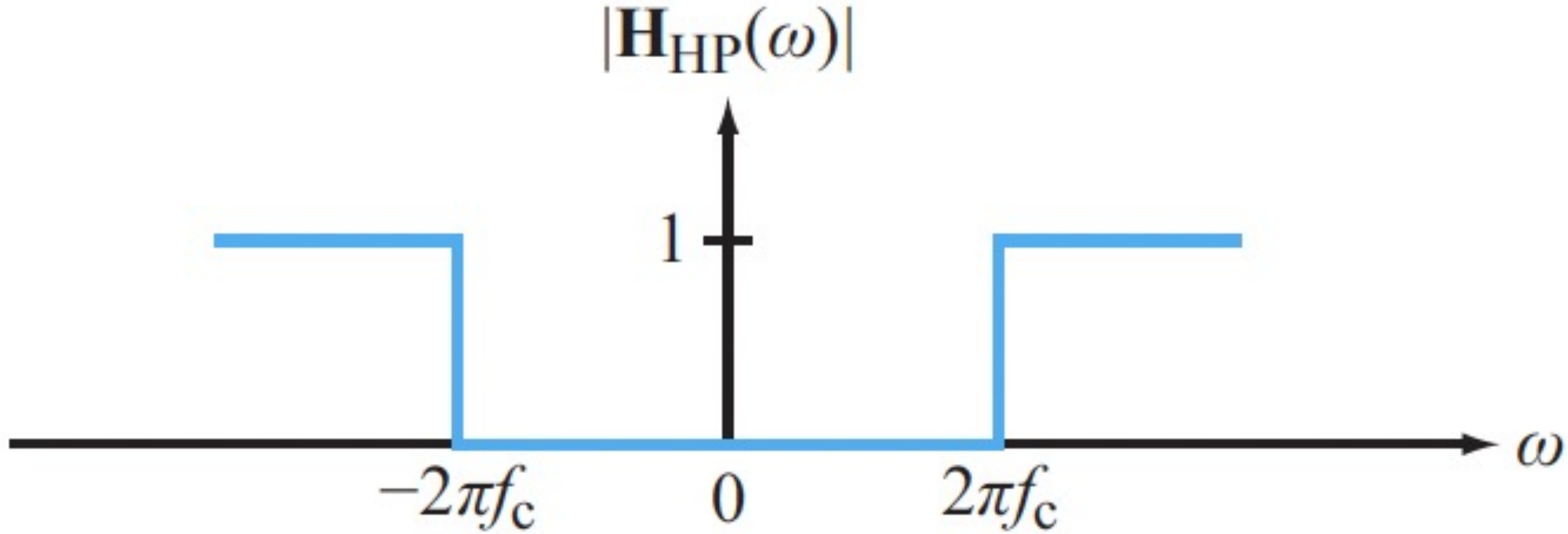
$$S_g = -20n \qquad \text{dB/decade}$$

- When $n = 1$, it reduces to the RC circuit
- Higher-order Butterworth filters offer better approximations to the ideal filter

$$M_{LP}(\omega) = \frac{|\mathbf{C}|}{\sqrt{\omega^{2n} + \omega_c^{2n}}} = \frac{|\mathbf{C}|}{\omega_c^n} \cdot \frac{1}{\sqrt{1 + (\omega/\omega_c)^{2n}}} \,.$$

(Butterworth lowpass filter)

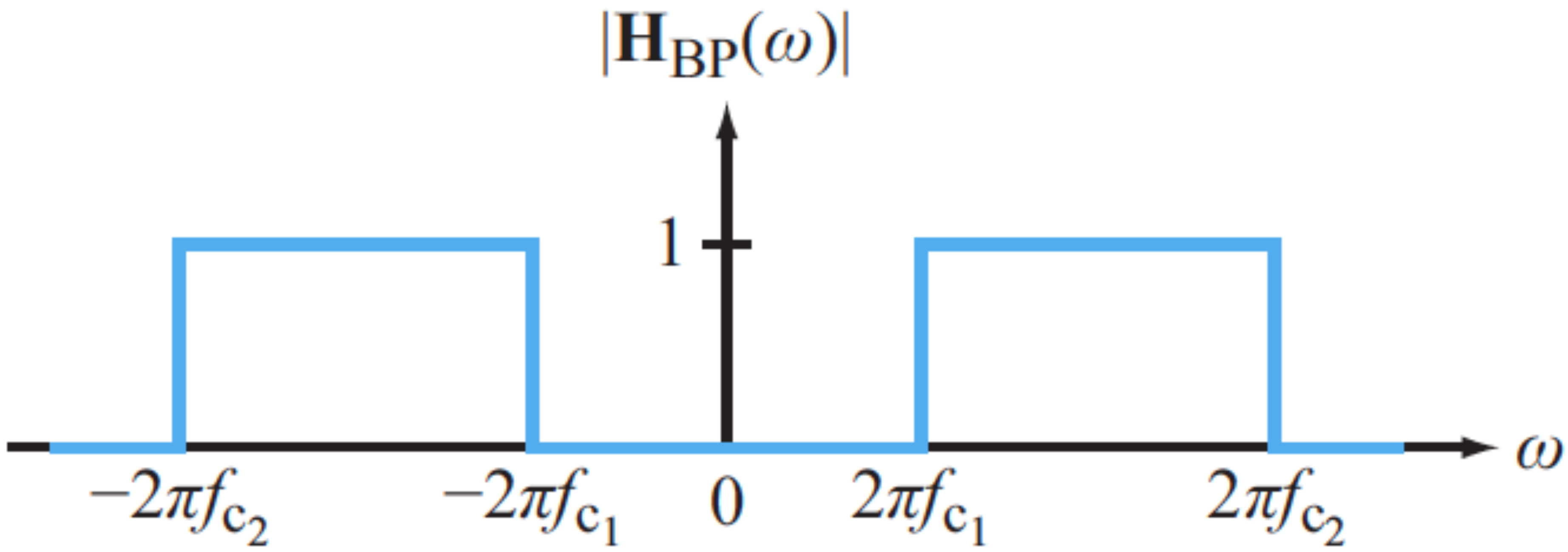
Brick-wall High-pass Filter



$$\mathbf{H}_{\mathrm{HP}}(\omega) = 1 - \mathbf{H}_{\mathrm{LP}}(\omega)$$

$$h_{\mathrm{HP}}(t) = \delta(t) - h_{\mathrm{LP}}(t) = \delta(t) - 2 f_c \,\mathrm{sinc}(2\pi f_c t)$$

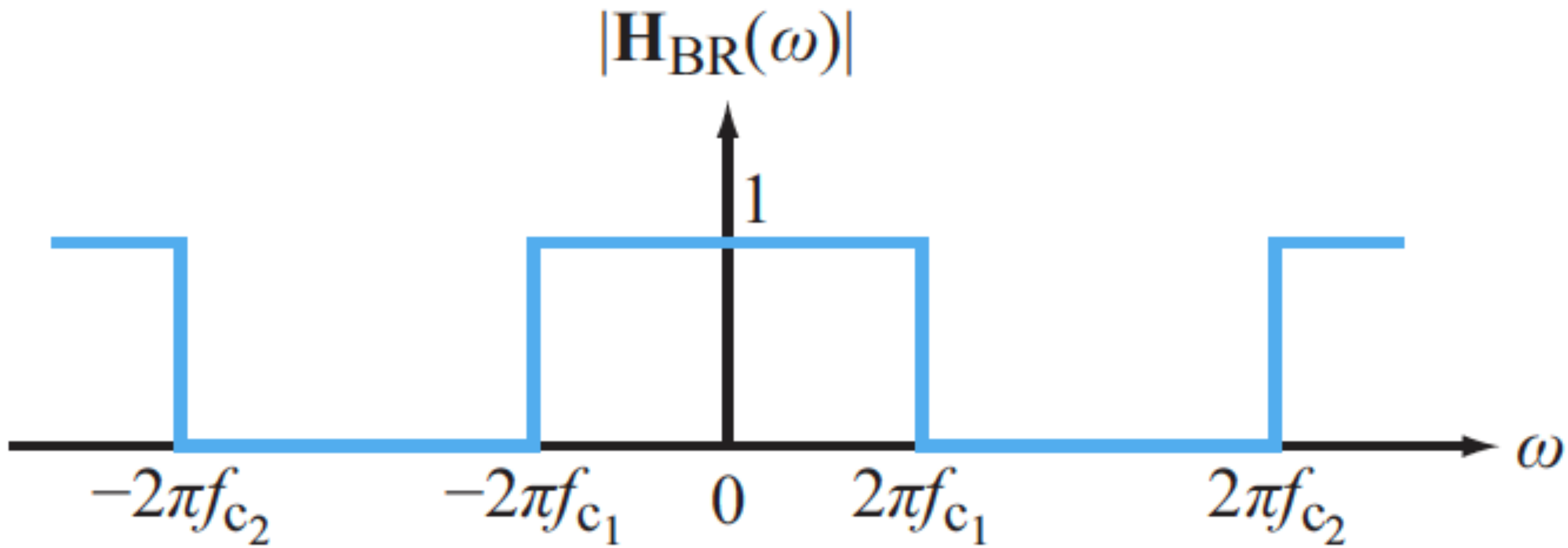
Brick-wall Band-pass Filter



$$h_{\mathrm{LP}}(t)\cos(\omega_0 t) \longleftrightarrow \frac{1}{2}[\mathbf{H}_{\mathrm{LP}}(\omega-\omega_0)+\mathbf{H}_{\mathrm{LP}}(\omega+\omega_0)]$$

$$h_{\mathrm{BP}}(t)=2(f_{c_2}-f_{c_1})\,\mathrm{sinc}[\pi(f_{c_2}-f_{c_1})t]\cos[\pi(f_{c_2}+f_{c_1})t]$$

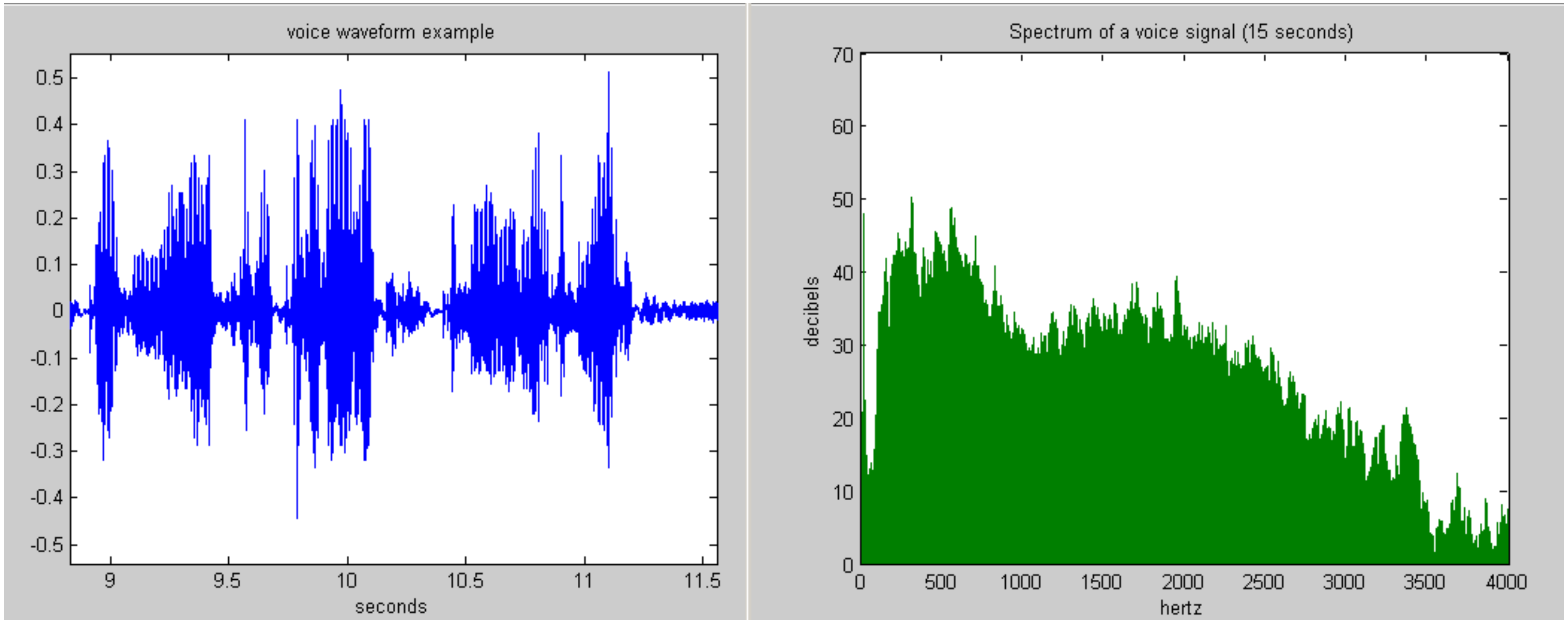
Brick-wall Band-reject Filter



$$\mathbf{H}_{\mathrm{BR}}(\omega) = 1 - \mathbf{H}_{\mathrm{BP}}(\omega)$$

$$h_{\mathrm{BR}}(t) = \delta(t) - h_{\mathrm{BP}}(t)$$

Typical Human Voice Spectrum



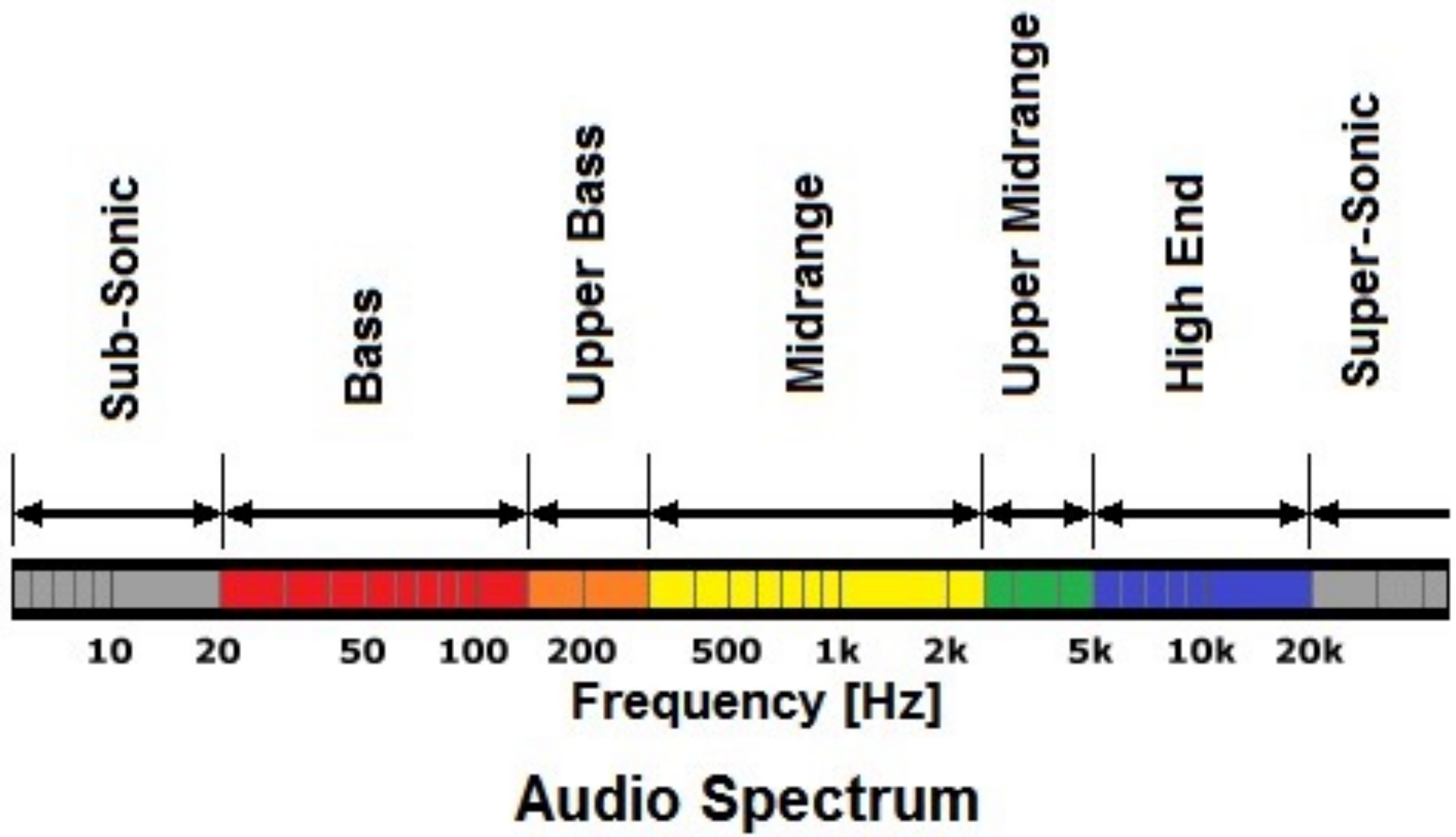
Male voice: Bulk of average power below 4kHz

Female voice: Bulk of average power below 6kHz

Ultimate Human Range of Hearing: 20Hz – 20kHz

Typical adult range of hearing: 20Hz – 10kHz

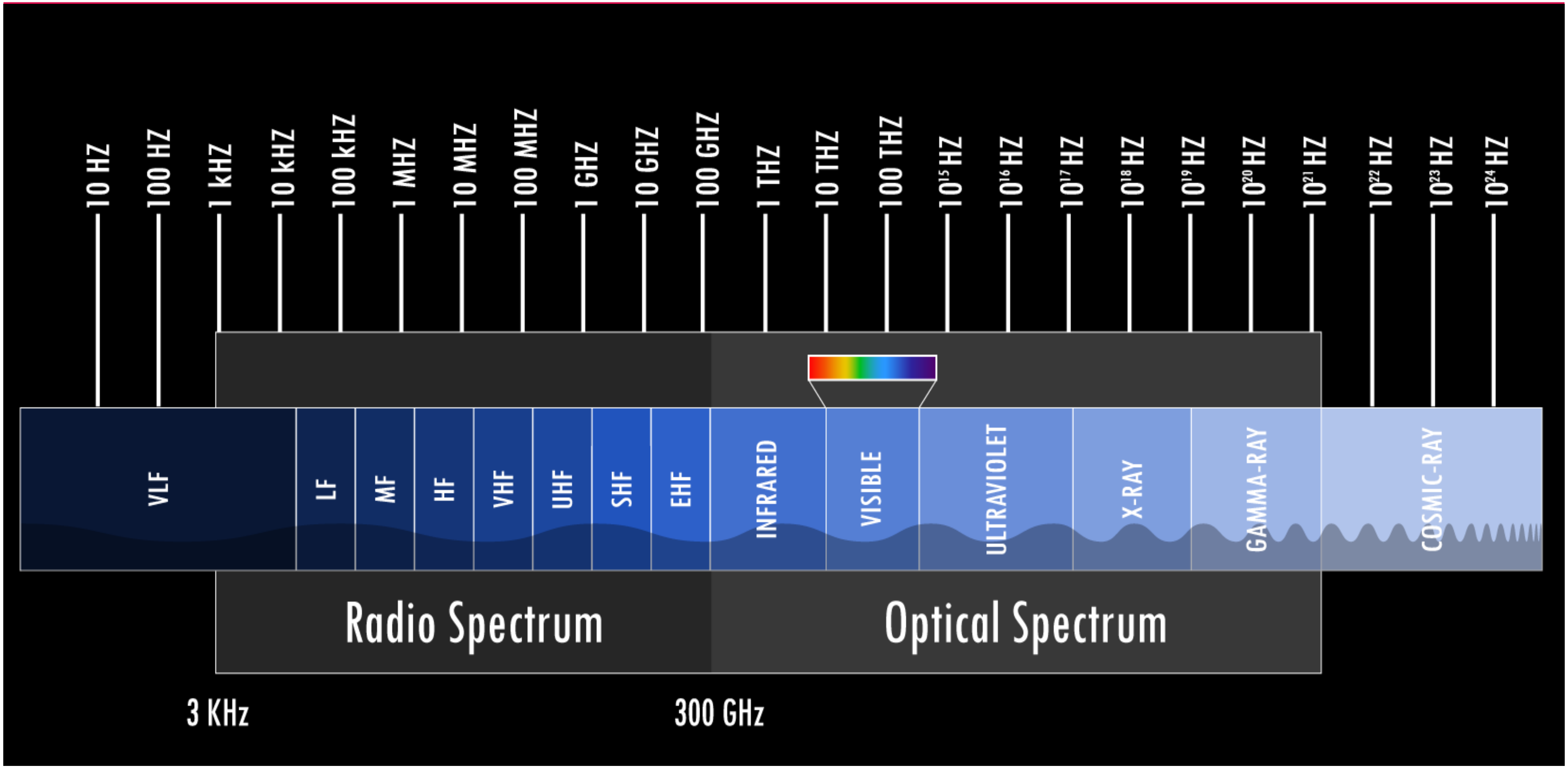
Human Audible Spectrum



Audio systems are designed to handle signals in the 20Hz – 20kHz range

https://www.audioreputation.com/audio-frequency-spectrum-explained/

The Electromagnetic Spectrum



UNITED STATES FREQUENCY ALLOCATIONS

THE RADIO SPECTRUM

RADIO SERVICES COLOR LEGEND

- AERONAUTICAL MOBILE
- AERONAUTICAL MOBILE SATELLITE
- AERONAUTICAL RADIONAVIGATION
- AMATEUR
- AMATEUR SATELLITE
- BROADCASTING
- BROADCASTING SATELLITE
- EARTH EXPLORATION SATELLITE
- FIXED
- FIXED SATELLITE
- INTER-SATELLITE
- LAND MOBILE
- LAND MOBILE SATELLITE
- MARITIME MOBILE
- MARITIME MOBILE SATELLITE
- MARITIME RADIONAVIGATION
- METEOROLOGICAL
- METEOROLOGICAL SATELLITE
- MOBILE
- MOBILE SATELLITE
- RADIO ASTRONOMY
- RADIODETERMINATION SATELLITE
- RADIOLOCATION
- RADIOLOCATION SATELLITE
- RADIONAVIGATION
- RADIONAVIGATION SATELLITE
- SPACE OPERATION
- SPACE RESEARCH
- STANDARD FREQUENCY AND TIME SIGNAL
- STANDARD FREQUENCY AND TIME SIGNAL SATELLITE

ACTIVITY CODE

- FEDERAL EXCLUSIVE
- FEDERAL/NON-FEDERAL SHARED
- NON-FEDERAL EXCLUSIVE

ALLOCATION USAGE DESIGNATION

SERVICE	EXAMPLE	DESCRIPTION
Primary	FIXED	Capital Letters
Secondary	Mobile	1st Capital with lower case letters

This chart is a graphic single-point-in-time portrayal of the Table of Frequency Allocations used by the FCC and NTIA. As such, it may not completely reflect all aspects, i.e. footnotes and recent changes made to the Table of Frequency Allocations. Therefore, for complete information, users should consult the Table to determine the current status of U.S. allocations.

U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management

JANUARY 2016

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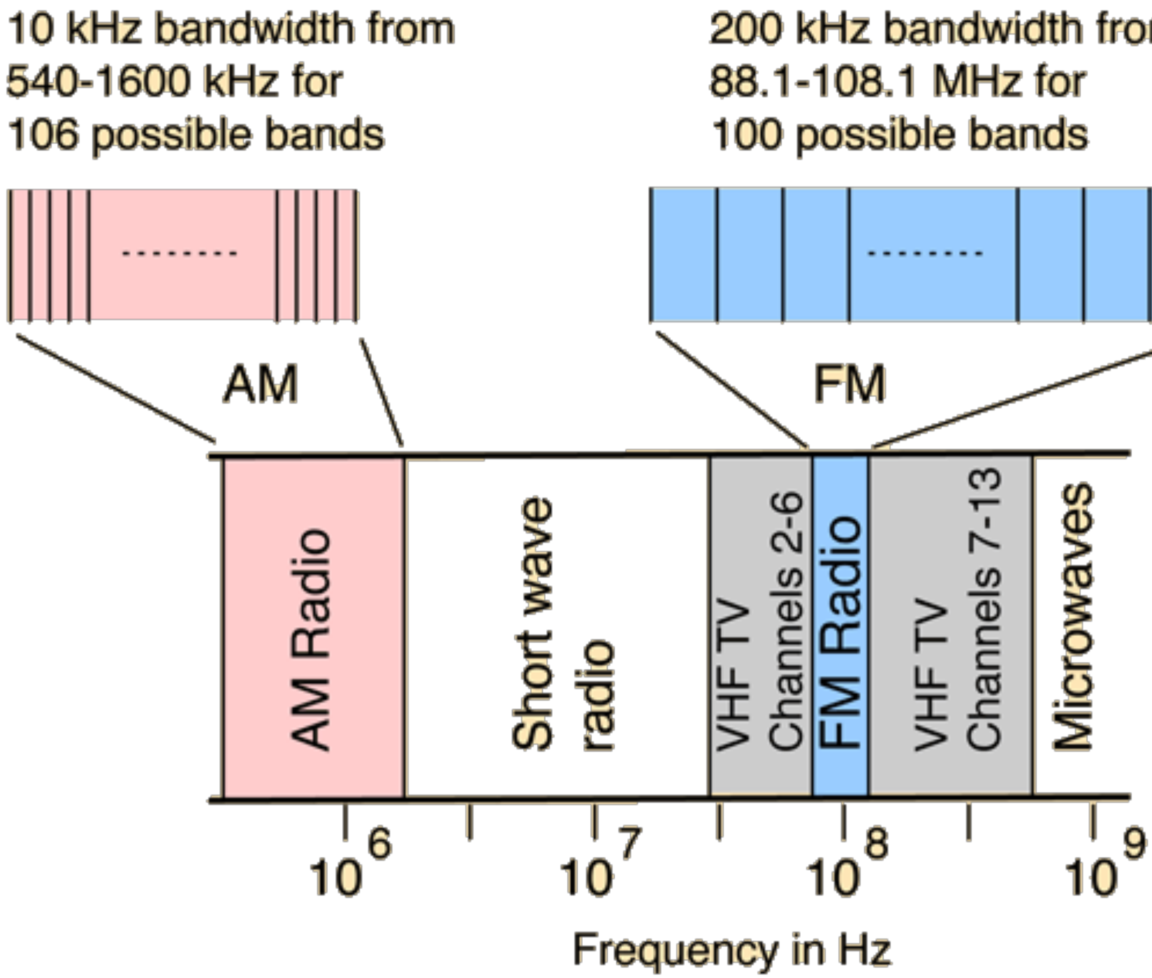
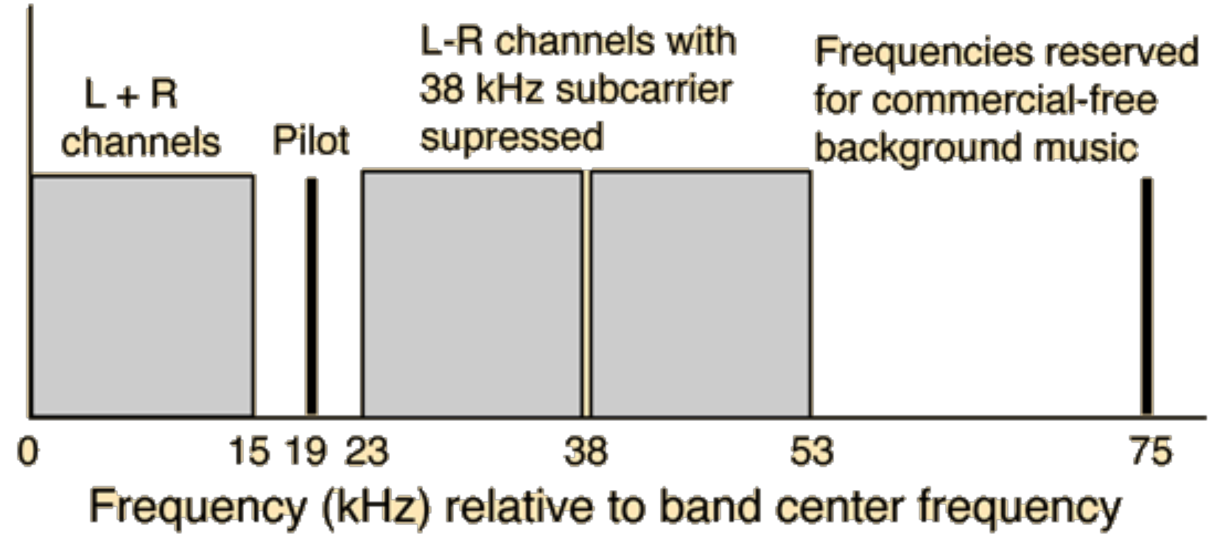
* EXCEPT AERONAUTICAL MOBILE (R)

** EXCEPT AERONAUTICAL MOBILE

PLEASE NOTE: THE SPACING ALLOTTED THE SERVICES IN THE SPECTRUM SEGMENTS SHOWN IS NOT PROPORTIONAL TO THE ACTUAL AMOUNT OF SPECTRUM OCCUPIED.

AM and FM Radio Bands





The FM Stereo Band provides for two audio channels, text, image, etc., leaving enough bandwidth for HD Radio (compressed audio), and other future developments.

From Analog Signals to Digital

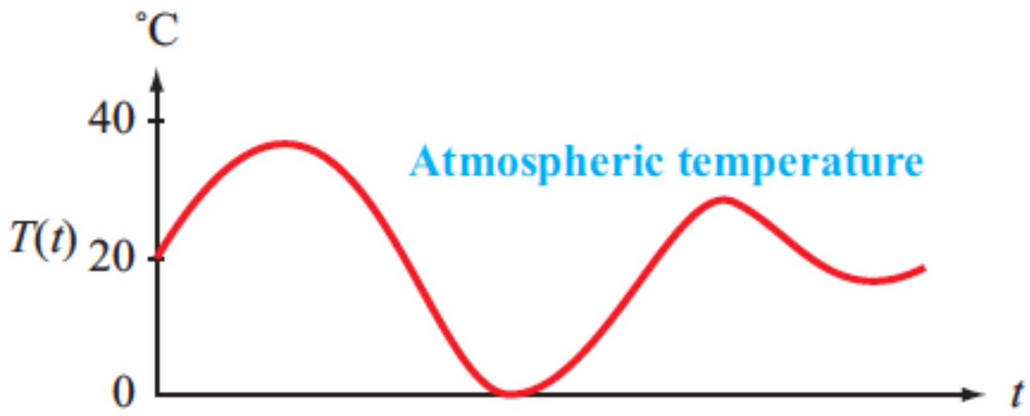
This course has been focused squarely on 100% analog (continuous time, continuous valued) signals and systems

- This is the necessary conceptual base

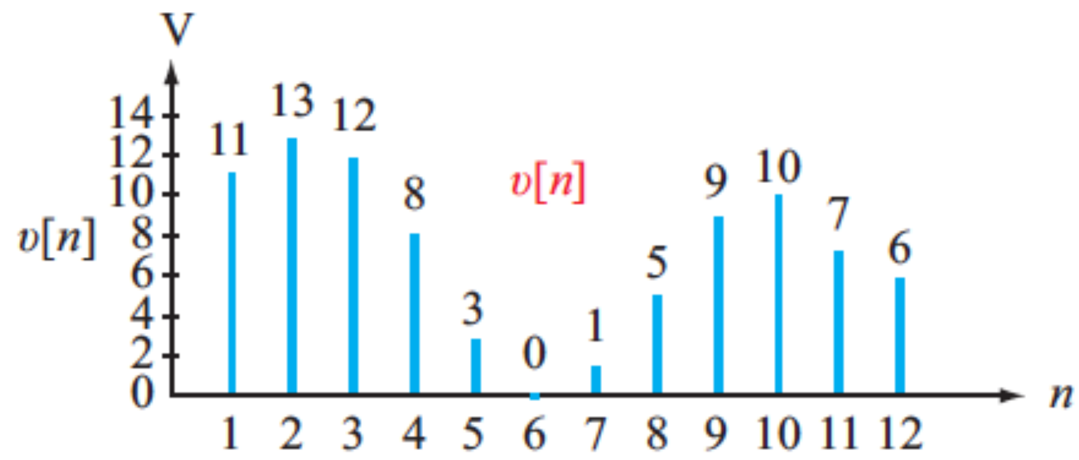
However, most signals and signal-processing systems in use today are digital

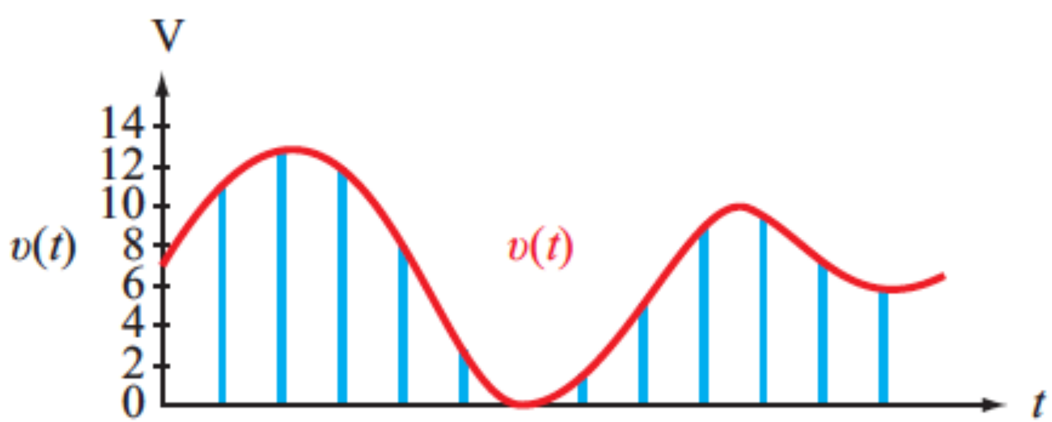
- Discrete-time
- Discretized values of signals
- Many benefits: digital systems are easier, cheaper, and more compact to implement (no bulky capacitors or inductors), flexible and easier to modify (e.g., software/firmware changes rather than hardware changes), can be less susceptible to noise and interference, can be stored and played back, made secure by encryption, etc.

Signal Sampling Basics

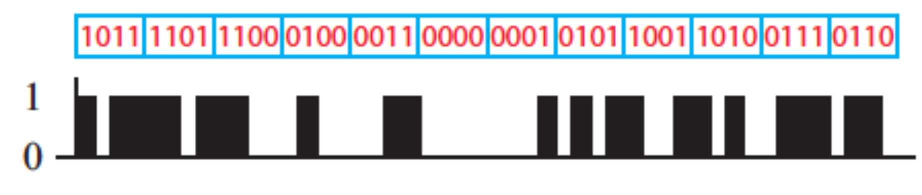


Continuous time
Continuous-valued amplitude





Discrete time
Continuous valued amplitude



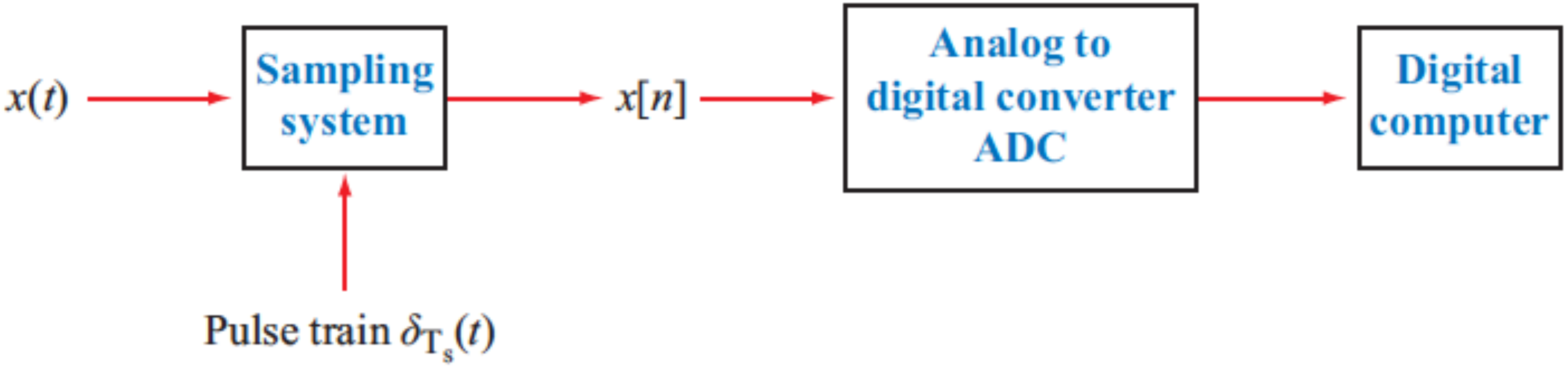
(d) Digital signal

Discrete time
Discrete-valued amplitude

Analog to Digital Conversion

Each sampled point can be quantized to a discrete *n*-bit number using an Analog to Digital Converter (ADC)

- An 8-bit ADC discretizes the signal to $2^8 = 256$ possible values
- Each added bit doubles the dynamic range, and therefore the precision (detail)



Opens endless possibilities to store, process, and transmit signals using digital hardware and (flexible) software

A Digital to Analog converter (DAC) can reproduce the analog signal from the digital form

Two Crucial Questions

#1 Sampling Rate: How often (in time) does the signal need to be sampled?

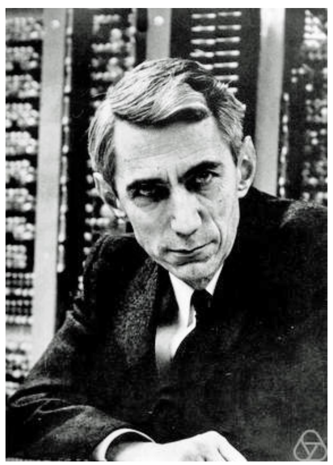
- If you sample too often, you have too much data to store/process
- If you sample insufficiently often, you are not representing the signal accurately

#2 Amplitude Discretization: How many bits of precision do we need for each sample?

- If we use too many bits, it can mean wasted storage, computation, and communication
- If we use too few bits, we are representing the signal crudely or inaccurately (the quantization noise is unacceptably high)

#1 Big Claim re: Sampling Rate

Harry Nyquist
(1889 - 1976,
Swedish American, Texan)

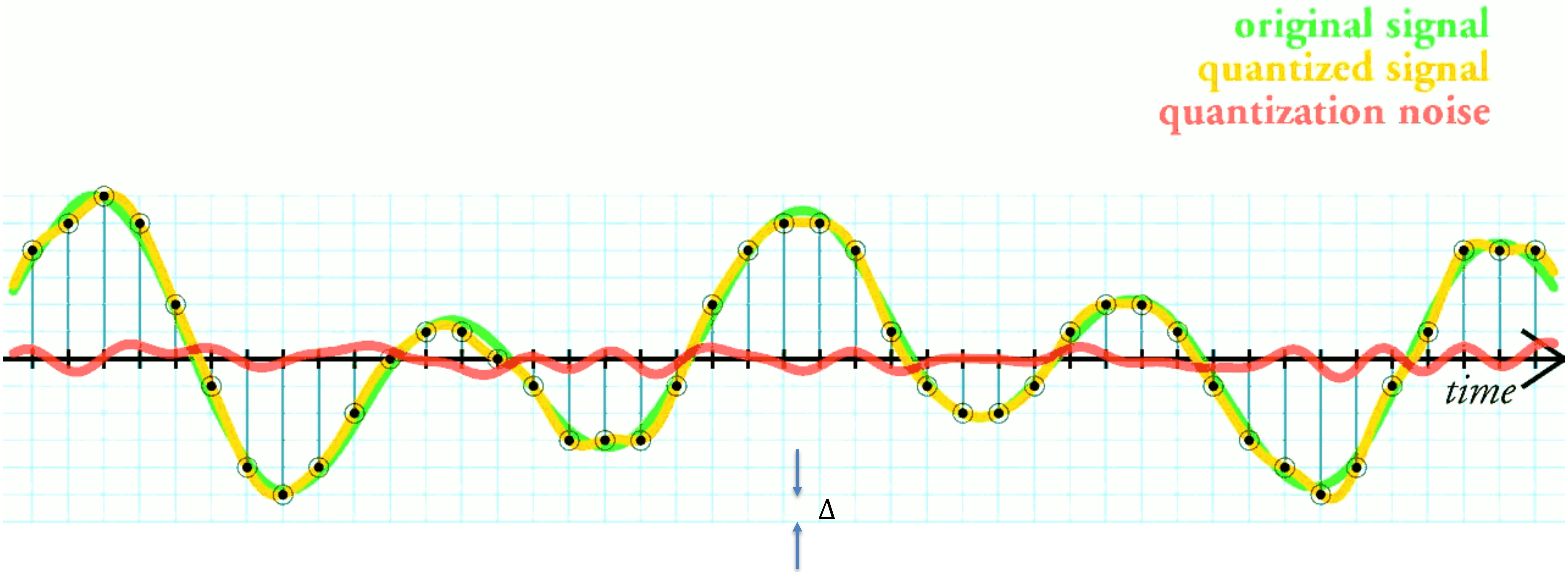


Claude Elwood Shannon
(1916 – 2001, USA)

Sampling Theorem

- Let $x(t)$ be a real-valued, continuous-time, lowpass signal *bandlimited* to a maximum frequency of B Hz.
- Let $x[n] = x(nT_s)$ be the sequence of numbers obtained by *sampling* $x(t)$ at a sampling rate of f_s samples per second, that is, every $T_s = 1/f_s$ seconds.
- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.

#2: Quantization Error (Noise)



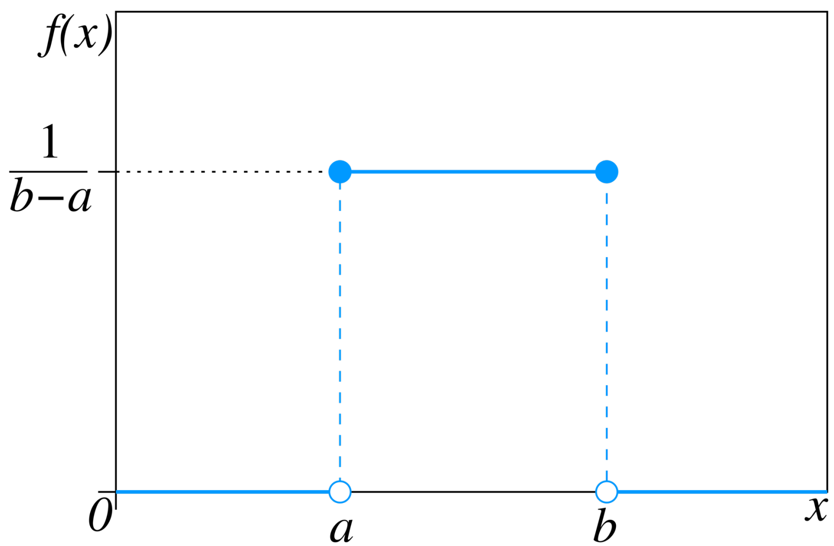
$$\text{Quantization Noise Power} = \frac{\Delta^2}{12}$$

We use sufficient bits to achieve the desired signal-to-noise power ratio

Quantization Noise Power

$$\text{Quantization Noise Power} = \frac{\Delta^2}{12}$$

- **This formula comes from the variance of the uniform distribution in statistics**
  - **The noise is assumed to be uniformly distributed with a mean of 0 and range Δ**



$$Mean = \frac{1}{2}(a+b)$$

$$Variance = \frac{1}{12}(b-a)^2$$

A Basic Pre-requisite

A signal must be band limited (finite bandwidth)

- Let $x(t)$ be a real-valued, continuous-time, lowpass signal *bandlimited* to a maximum frequency of B Hz.

A band-limited signal has a spectrum with a finite overall bandwidth *B*

If the signal is not band limited, we cannot sample it

No frequencies higher than *B*

This places an upper bound on how fast the signal can change

#1 Sampling Rate

- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.

This is neat: Shannon gives us a specific answer on just how often to sample!

What is even neater is that we can recover the original signal without any loss from sampling if

$$f_S > 2B$$

#1 Sampling Rate

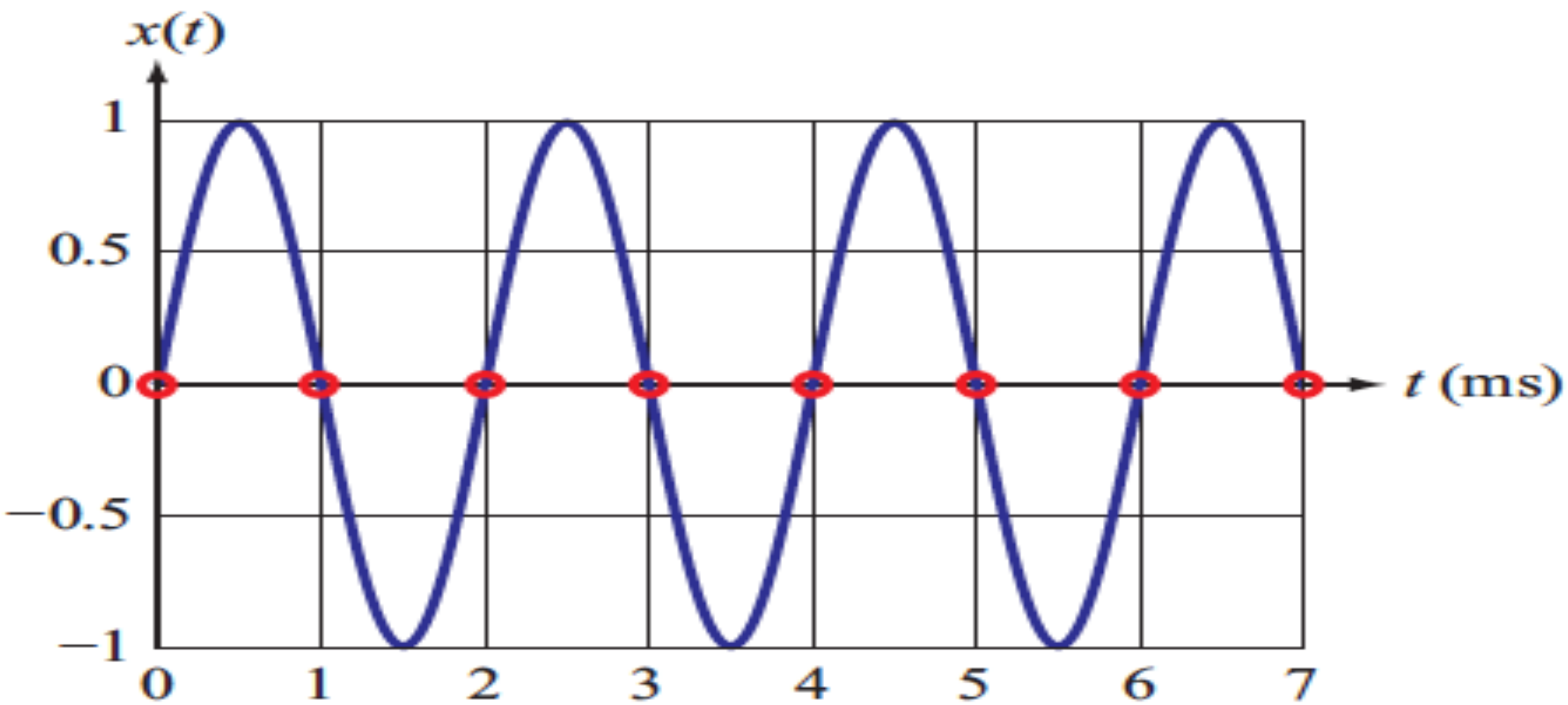
- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.

2*B* is commonly referred to as the *Nyquist Rate*

Harry Nyquist
(1889 - 1976,
Swedish American, Texan)

Can we sample at $f_S = 2B$?

- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.



YIKES!! WE MISSED IT TOTALLY

500Hz signal sampled at 2B = 1000Hz (not enough!)

The sampling rate must be greater than 2B!!

$$f_S = 2B$$

is not good enough

Let's Derive the Sampling Theorem !

Fourier transform concepts are used to derive the sampling theorem

- **Since the sampling theorem was stated in terms of the frequency f rather than angular frequency ω, it is convenient to work in the f space**

$$\omega = 2\pi f$$

- **All the Fourier properties are unchanged, for example:**

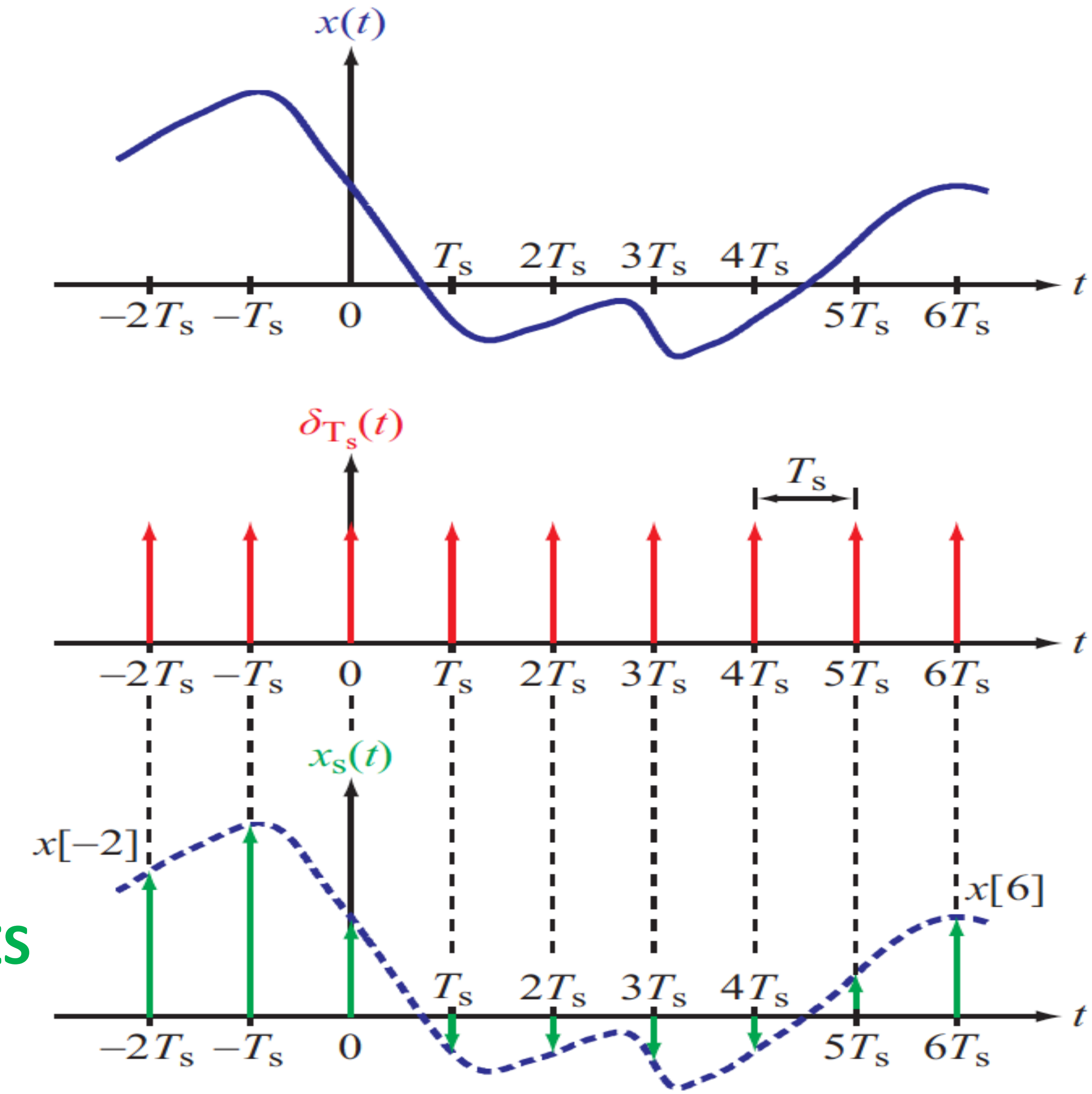
$$x_1(t)\, x_2(t) \quad \longleftrightarrow \quad \mathbf{X}_1(f) * \mathbf{X}_2(f)$$

Signal Sampling

GIVEN SIGNAL

IMPULSE TRAIN

SIGNAL SAMPLES



$$\delta_{T_S}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_S)$$

$$x_S(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_S)$$

$$= \sum_{n=-\infty}^{\infty} x(nT_S)\delta(t - nT_S)$$

$$= \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_S)$$

Signal Sampling

$$\delta_{T_S}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_S)$$ **PULSE TRAIN**

$$x_S(t) = x(t)\delta_{T_S}(t)$$

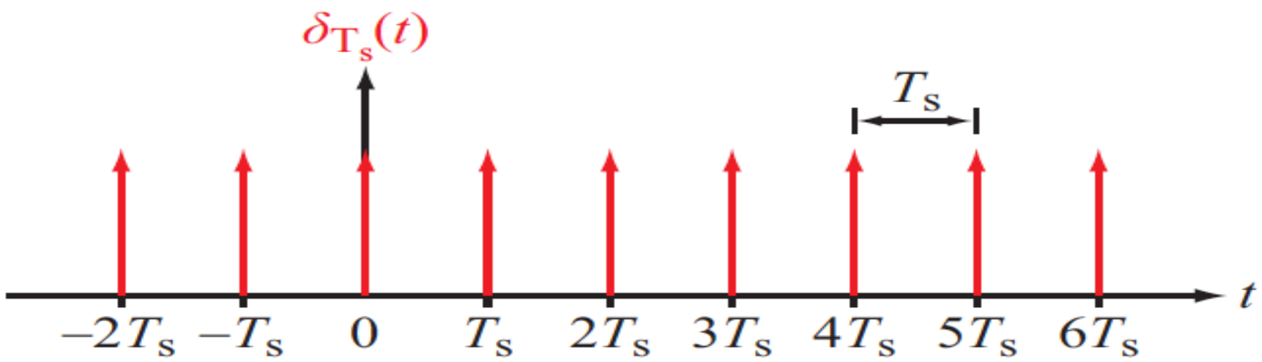
This product of signals in the time domain becomes a convolution in the frequency domain

$$x_{\mathrm{S}}(t) = x(t)\ \delta_{\mathrm{T_S}}(t) \quad \longleftrightarrow \quad \mathbf{X}_{\mathrm{S}}(f) = \mathbf{X}(f) * \mathbf{\Lambda}(f)$$

FOURIER TRANSFORM OF THE PULSE TRAIN

Fourier Transform of a Pulse Train

$$\delta_{T_S}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_S)$$



Watch: We cannot calculate the Fourier transform of this signal directly by integration because it has an infinite number of impulse discontinuities and is not absolutely integrable!

- **We need a different strategy**
  - **Basic idea: The pulse train is a periodic signal, so we can start off by calculating its Fourier Series instead, and this is valid (signal meets its requirements for one period)**

Recap: Exponential form of Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$$

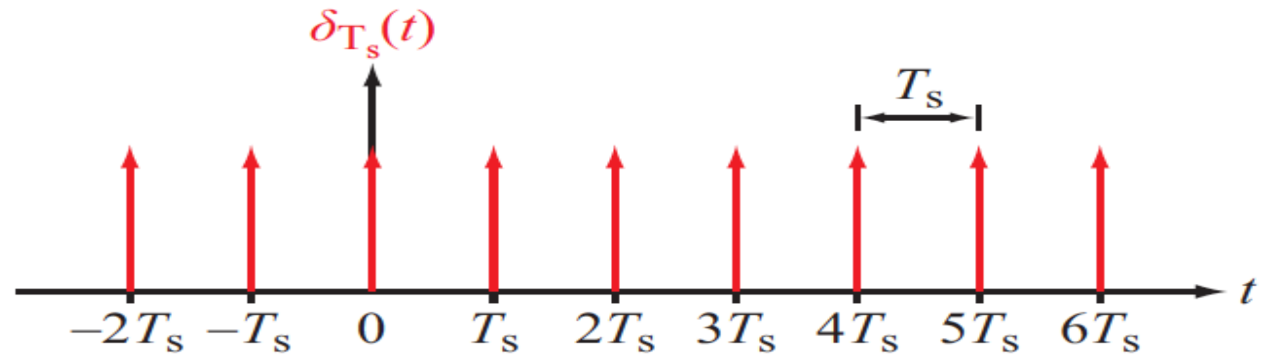
$$\mathbf{x}_n = |\mathbf{x}_n| e^{j\phi_n}; \;\; \mathbf{x}_{-n} = \mathbf{x}_n^*; \;\; \phi_{-n} = -\phi_n$$

$$|\mathbf{x}_n| = c_n/2; \;\; x_0 = c_0$$

$$\mathbf{x}_n = \frac{1}{T_0} \int_0^{T_0} x(t)\, e^{-jn\omega_0 t}\, dt$$

Fourier Series of the Pulse Train

$$\delta_{T_S}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_S) = \sum_{n=-\infty}^{\infty} \mathbf{d}_n e^{j2\pi n f_S t}$$



$$\mathbf{d}_n = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta_{T_s}(t)\, e^{-j2\pi n f_s t}\, dt = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t)\, e^{-j2\pi n f_s t}\, dt = \frac{1}{T_s} = f_s$$

$$\Rightarrow \delta_{T_S}(t) = f_S \sum_{n=-\infty}^{\infty} e^{j2\pi n f_S t}$$

Fourier Series of the Pulse Train

$$\delta_{T_S}(t) = f_S \sum_{n=-\infty}^{\infty} e^{j2\pi n f_S t}$$ **FOURIER SERIES**

$$e^{j2\pi n f_s t} \leftrightarrow \delta(f - n f_s)$$ **FOURIER TRANSFORM**

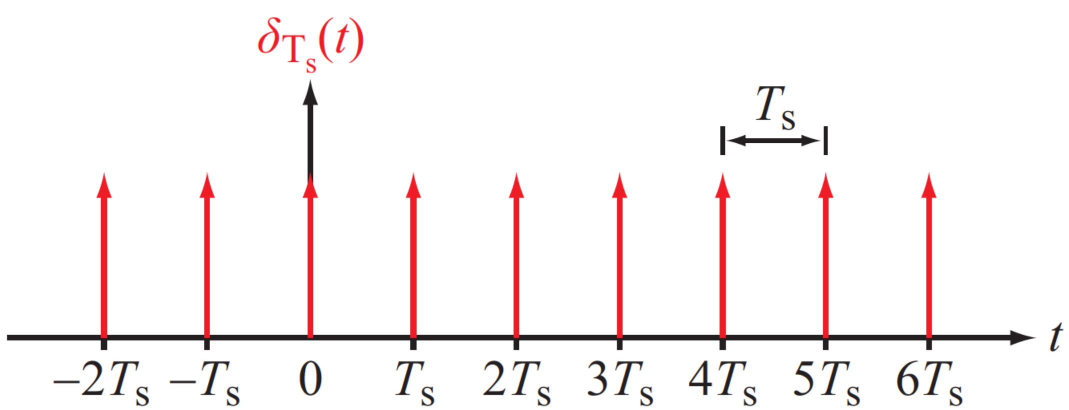
$$\Lambda(f) = \mathcal{F}[\delta_{T_s}(t)] = f_s \sum_{n=-\infty}^{\infty} \delta(f - n f_s)$$

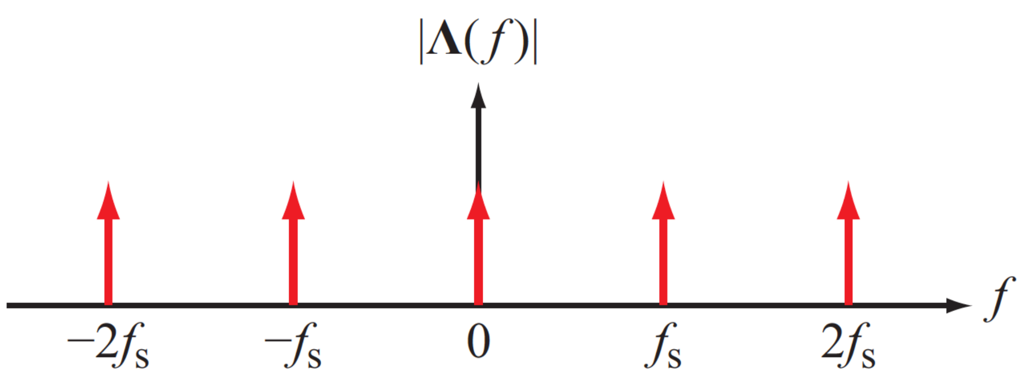
<u>THIS IS FASCINATING</u>:

THE FOURIER TRANSFORM OF THE PULSE TRAIN IS A PULSE TRAIN IN THE FREQUENCY DOMAIN

Fourier Series of the Pulse Train

$$\Lambda(f) = \mathcal{F}[\delta_{T_s}(t)] = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

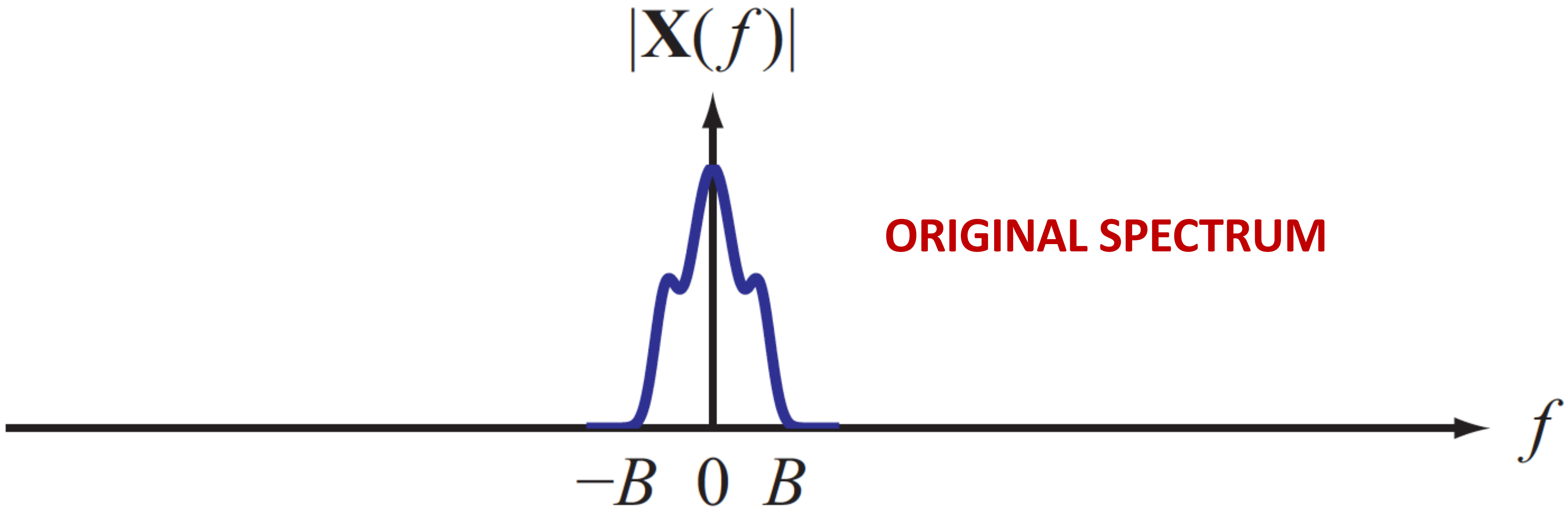




FASCINATING:

THE FOURIER TRANSFORM OF THE PULSE TRAIN IS A PULSE TRAIN, IN THE FREQUENCY DOMAIN

Fourier Transform of the Signal

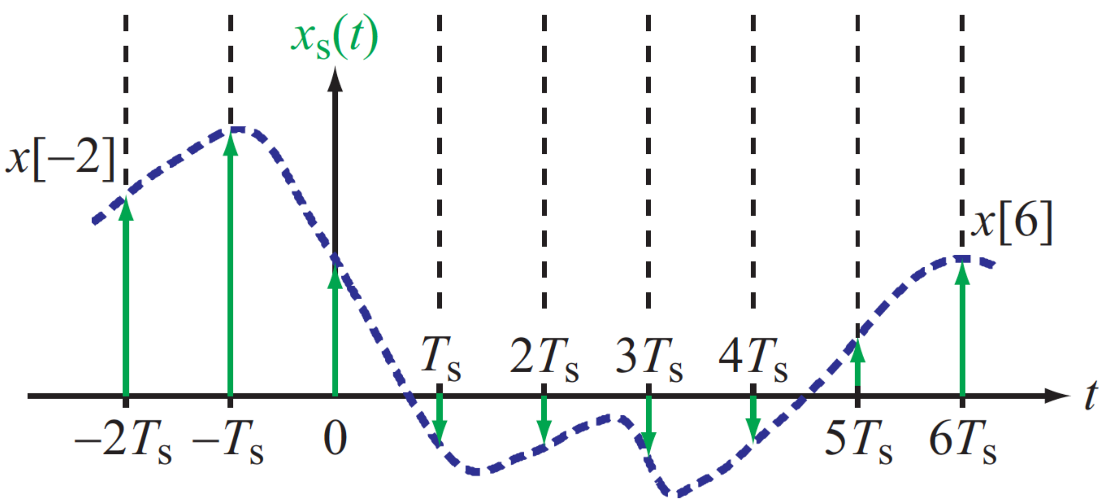


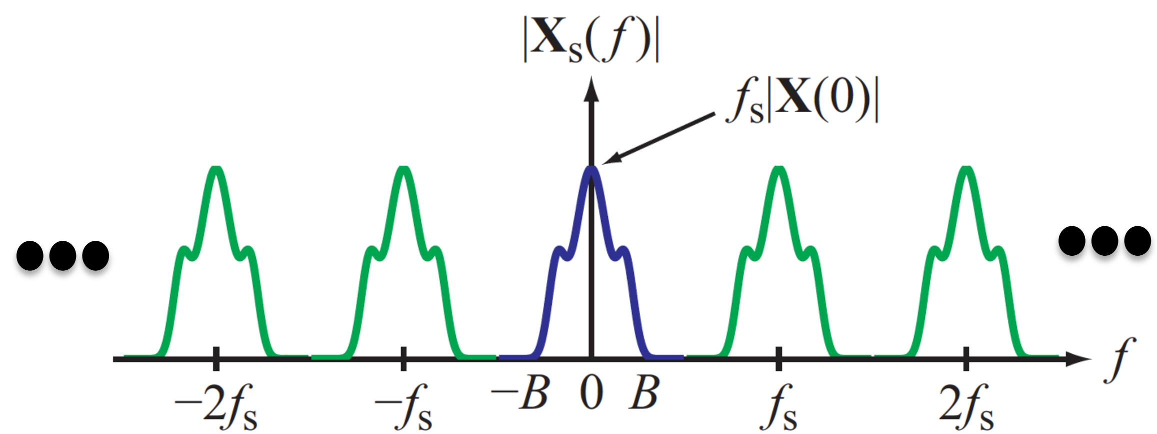
ORIGINAL SPECTRUM

Fourier Series of the Sampled Signal

CONVOLVE

$$\mathbf{X}_s(f) = \mathbf{X}(f) * f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \mathbf{X}(f - nf_s)$$





WE CAN NOW EXAMINE THE EFFECT OF VARYING THE SAMPLING FREQUENCY

Summary

We introduced the sampling theorem today

- Let $x(t)$ be a real-valued, continuous-time, lowpass signal *bandlimited* to a maximum frequency of B Hz.
- Let $x[n] = x(nT_s)$ be the sequence of numbers obtained by *sampling* $x(t)$ at a sampling rate of f_s samples per second, that is, every $T_s = 1/f_s$ seconds.
- Then $x(t)$ can be *uniquely* reconstructed from its samples $x[n]$ if and only if $f_s > 2B$. The sampling rate must exceed double the bandwidth.

Putting it to use requires additional work (next class!)